

Package ‘accmv’

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Type Package

Title Inference with Available Complete-Case Missing Values

Version 0.1.1

Description Implements inverse probability weighted, regression adjustment, and multiply robust estimators under the available complete-case missing value assumption of Cheng, Chen, Smith, and Zhao (2022) [<doi:10.48550/arXiv.2207.02289>](https://doi.org/10.48550/arXiv.2207.02289). Supports one or two primary variables, exponential-tilt sensitivity analysis, regression weights, and nonparametric bootstrap confidence intervals.

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Suggests testthat (>= 3.0.0)

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URL <https://github.com/mathcg/ACCMV>, <https://arxiv.org/abs/2207.02289>

BugReports <https://github.com/mathcg/ACCMV/issues>

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accmv-package	<i>Inference with Available Complete-Case Missing Values</i>
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Description

Implements inverse-probability-weighted, regression-adjustment, and multiply-robust estimators under the available complete-case missing value assumption of Cheng, Chen, Smith, and Zhao (2022).

References

Cheng G, Chen Y-C, Smith MA, Zhao Y-Q (2022). *Handling Nonmonotone Missing Data with Available Complete-Case Missing Value Assumption*. <https://arxiv.org/abs/2207.02289>.

accmv_data	<i>Prepare ACCMV Data</i>
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Description

Validates data and identifies observed-value patterns.

Usage

```
accmv_data(x, y)
single_data_preparation(x, y)
multiple_data_preparation(x, y)
```

Arguments

x	Numeric secondary-variable matrix with missing values represented by NA.
y	Numeric primary-variable vector or matrix.

Value

An object containing the data and stable missing-pattern encodings.

Examples

```
x <- matrix(c(1, NA, 2, 3, NA, 4), ncol = 2)
y <- c(1, NA, 2)
accmv_data(x, y)
```

bootstrap_accmv_single

Bootstrap ACCMV Estimators

Description

Performs a nonparametric row bootstrap, retrying draws lacking estimable pattern comparisons.

Usage

```
bootstrap_accmv_single(data, method = "mr", n_B = 999, seed = NULL, ...)
bootstrap_accmv_multiple(data, method = "mr", target = "average",
  threshold = NULL, n_B = 999, seed = NULL, delta = NULL)
single_bootstrap(data, method, n_B = 1000, ...)
multiple_bootstrap(data, method, n_B = 1000, ...)
bootstrap_regression(data, method, fm, n_B = 1000, ...)
```

Arguments

data	An object from accmv_data().
method	Estimator name or a function; for regression bootstrap, a model-fitting function.
n_B	Number of accepted bootstrap replicates.
seed	Optional random seed.
target	Built-in multiple-primary target.
threshold	Threshold for the indicator target.
delta	Optional exponential-tilt sensitivity parameter.
...	Additional estimator arguments.
fm	Regression formula using primary-variable columns.

Value

A numeric vector of bootstrap estimates.

estimate_accmv_single *ACCMV Estimators*

Description

Fits the estimators described in Cheng et al. (2022).

Usage

```

estimate_accmv_single(x, y, method = "mr", fun = identity,
  binary = FALSE, delta = NULL, n_boot = 0, level = 0.95, seed = NULL)
estimate_accmv_multiple(x, y, method = "mr", target = "average",
  threshold = NULL, delta = NULL, n_boot = 0, level = 0.95, seed = NULL)
single_regression_adjustment(data, fun = identity, binary = FALSE)
single_ipw(data, fun = identity, delta = NULL)
single_ipw_sensitivity(data, delta, fun = identity)
single_multiply_robust(data, fun = identity, binary = FALSE)
multiple_ipw(data, fun = NULL, target = "average", threshold = NULL,
  delta = NULL)
multiple_ipw_sensitivity(data, delta, fun = NULL, target = "average",
  threshold = NULL)
multiple_ra_average(data)
multiple_ra_product(data)
multiple_ra_indicator(data, a)
multiple_mr_average(data)
multiple_mr_product(data)
multiple_mr_indicator(data, a)
accmv_ipw_weights(data)
ipw_regression(data)

```

Arguments

x	Numeric secondary-variable matrix.
y	Numeric primary-variable vector or two-column matrix.
method	One of "ipw", "ra", or "mr".
fun	Transformation defining the estimand.
binary	Whether the transformed outcome is binary.
delta	Optional exponential-tilt sensitivity parameter.
n_boot	Number of bootstrap replicates; zero disables bootstrap.
level	Confidence level.
seed	Optional random seed.
target	One of "average", "product", or "indicator".
threshold	Threshold for the indicator target.
data	An object from accmv_data().
a	Threshold for the indicator target.

Value

High-level functions return an `accmv_result`; low-level functions return an estimate or weights.

Examples

```
sim <- simulate_accmv_single(500, seed = 1)
estimate_accmv_single(sim$x, sim$y, method = "ra")

sim2 <- simulate_accmv_multiple(500, seed = 2)
estimate_accmv_multiple(sim2$x, sim2$y, method = "mr", target = "product")
```

fit_accmv_regression	<i>Fit an ACCMV-Weighted Marginal Linear Model</i>
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Description

Fits a weighted least-squares marginal model using the ACCMV IPW weights from Section 5.

Usage

```
fit_accmv_regression(x, y, response = 2, predictors = 1,
  n_boot = 0, level = 0.95, seed = NULL)
```

Arguments

x	Numeric secondary-variable matrix.
y	Numeric primary-variable matrix.
response	One-based response column in y.
predictors	One-based predictor columns in y.
n_boot	Number of bootstrap replicates; zero disables bootstrap.
level	Confidence level.
seed	Optional bootstrap seed.

Value

An accmv_regression_result with coefficients, weights, and optional bootstrap inference.

Examples

```
sim <- simulate_accmv_regression(500, seed = 3)
fit_accmv_regression(sim$x, sim$y)
```

```
print.accmv_regression_result
```

Print an ACCMV Regression Result

Description

Prints the ACCMV-weighted regression coefficients.

Usage

```
## S3 method for class 'accmv_regression_result'
print(x, ...)
```

Arguments

x	An accmv_regression_result object.
...	Unused.

Value

The input object, invisibly.

```
print.accmv_result
```

Print an ACCMV Result

Description

Prints the method, target, point estimate, and bootstrap standard error when available.

Usage

```
## S3 method for class 'accmv_result'
print(x, ...)
```

Arguments

x	An accmv_result object.
...	Unused.

Value

The input object, invisibly.

simulate_accmv_single *Paper Simulation Designs*

Description

Generates data from Sections 7.1 and 7.2 of the ACCMV paper.

Usage

```
simulate_accmv_single(n = 2000, seed = NULL)
simulate_accmv_multiple(n = 2000, seed = NULL)
simulate_accmv_regression(n = 2000, seed = NULL)
```

Arguments

n	Sample size.
seed	Optional random seed.

Value

A list containing x and y.

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